



TSX & TSXV Level 1 TMX QuantumFeed Business Message Specifications

Document Revision: 2.6.0
Date of Issue: August 10, 2021

Disclaimer and Rights Granted

The information contained in this document and any other information or documentation related to the TMX QuantumFeed™ specification including all past, present and future versions thereof and associated message formats (collectively referred to as the "TMX QuantumFeed Specification") are provided "as is", and neither TSX Inc. ("TSX") nor any of its affiliates makes any representation or warranty, express or implied, as to the contents of the TMX QuantumFeed Specification (or the results obtained by using it) and each such person specifically disclaims any representation, warranty or condition of accuracy, completeness, merchantability, fitness for a particular purpose or that it is error-free. By using the TMX QuantumFeed Specification, you are agreeing to assume the entire risk associated with its use.

TSX and its affiliates shall have no liability for damages of any kind arising in any manner out of or in connection with your use of, or your inability to use, the TMX QuantumFeed Specification, whether direct, indirect, incidental, special or consequential (including, without limitation, loss of data, loss of use, third-party claims, lost profits, lost revenues and strict liability), arising under contract or otherwise, whether or not TSX has been advised of, or otherwise might have anticipated the possibility of, such damages.

The information contained in the TMX QuantumFeed Specification is proprietary and confidential information belonging to TSX and its licensors or service providers, as applicable. Copyright and trademark rights and any other intellectual property rights in the TMX QuantumFeed Specification belong to TSX (or its licensors or service providers, as applicable). Your permitted use of the TMX QuantumFeed Specification, in whole or in part, is limited to the non-exclusive, non-transferable, revocable, non-assignable, personal right for you only to build a connection between your systems and TSX's market information system. Any uses not specifically set out herein are prohibited. TSX may revoke this right at any time, on notice to you, and you agree to promptly comply with such notice. Do not reproduce, disclose, distribute, publish, sell, modify, commercialize or make any part of the TMX QuantumFeed Specification available, in whole or in part, to any person including to any of your affiliates, service providers or any other related party without TSX's prior written consent. If you do not agree with, or do not wish to comply with, any of the above, please notify TSX and, in accordance with TSX's instructions, either return or destroy any copies or version of the TMX QuantumFeed Specification in your possession.

This Notice, Disclaimer and Rights Granted sets out the terms and conditions on which you are permitted to use the TMX QuantumFeed Specification. This Notice, Disclaimer and Rights Granted may be changed from time to time by TSX in its discretion. It is your obligation to check and comply at all times with the terms applicable to the most current version of the TMX QuantumFeed Specification.

© TSX Inc. 2010 – 2020. All rights reserved.

Contact Information

For questions related to TMX QuantumFeed, contact TMX Market Operations at 416.947.4705 or marketops@tmx.com.

Contents

1 Introduction	4
1.1 Intended Audience	4
1.2 Scope	4
2 Business Messages Overview	5
2.1 Business message types	5
2.2 Message data types	6
2.3 Message field definitions	6
3 Business Message Formats	7
3.1 Message structure	7
3.2 Symbol Status	8
3.3 Trade	9
3.4 Trade Cancelled	10
3.5 MOC Imbalance	11
3.6 Stock Status	11
3.7 Quote	13
4 Field Definitions	14
A	14
B	14
C	15
F	17
I	17
L	19
M	19
O	21
P	22
R	24
S	24
T	25
V	26
Appendix A Acronyms and Definitions	27
Appendix B Revision History	28

1 Introduction

This document is the *TSX and TSXV Level 1 QuantumFeed Business Message Specifications*.

The Level 1 TMX QuantumFeed™ Business messages provide:

- Trades
- Quotes
- Symbol and stock status

1.1 Intended Audience

The intended audience of this specification consists of business analysts and programmer analysts.

1.2 Scope

The scope of this document is limited to describing the *Business messages* disseminated on the TSX and TSXV Level 1 TMX QuantumFeed. The document provides:

- The structure of each type of business message
- Fields that constitute those messages
- The data types and formats to which the fields comply
- More details on processing the information contained in those fields

To design and implement systems that can receive and process Level 1 QuantumFeed, this document must be used with the following documents:

- *TMX QuantumFeed Service Access Guide*
- *TMX eXtreme Message Transfer Protocol Specification*

2 Business Messages Overview

Business messages disseminated on TSX and TSXV Level 1 TMX QuantumFeed contain business-related information; for example: information on trades, information on quotes, and so on. A Business message is composed of two sections: a *message header* and the *message body*.

- The **message header** is common to all Business messages and provides information that is required to process the given message.
- The **message body** contains the business information.

2.1 Business message types

TSX and TSXV Level 1 TMX QuantumFeed contains the following types of Business messages.

Message Name	Message Type		Description
	ASCII	HEX	
Symbol Status	J	0x4A	Symbol Status messages contain information regarding equity, debenture, mutual fund, or trading Instruments for the current trading day for both TSX and TSXV markets. One message is disseminated for each valid symbol at the beginning of each trading day.
Trade	s	0x73	A Trade message is produced each time a trade occurs.
Trade Cancelled	t	0x74	A Trade Cancelled message is produced each time a trade is cancelled.
MOC Imbalance	u	0x75	A MOC Imbalance message is disseminated for every MOC-eligible stock every 10 seconds starting from the beginning of the MOC Imbalance session at 3:50 p.m., or as notified by TSX Equities Trading. The message will also be disseminated at 4:00 p.m. for closing delayed stocks.
Stock Status	v	0x76	A Stock Status message is disseminated in response to a change in stock status from the Trading Engine. It contains information regarding the current state of the stock, MOC Price movement delays, and so on.
Quote	w	0x77	A Quote message contains the latest Bid and Ask information for a symbol.

2.2 Message data types

Each field that is contained in the Business messages is formatted according to one of the following data types:

Data Type	Description
Alphanumeric	Alphanumeric fields are left-justified ASCII fields (displayable characters from hex 0x20 to hex 0x7E) that are space padded (hex 0x20) on the right and are sized to their corresponding lengths. Example: Consider a field with length <i>10 bytes</i>, data type <i>Alphanumeric</i> and which must contain the information “CASH”. In this scenario, the field contains the value “CASH ”. The information is left justified and is space (“ ”) padded on the right. The information and the padding (if required) make the field length 10.
Binary	Binary Level 1 QuantumFeed message fields are Little Endian encoded. Length of the binary fields is one of the following: • 1-Byte Binary • 2-Byte Binary • 4-Byte Binary • 8-Byte Binary Default Value: Null (binary value 0)

2.3 Message field definitions

For descriptions of fields that make up the Business messages, refer to “Field Definitions” on page 14.

3 Business Message Formats

This section contains the structure of each Business message disseminated on TSX and TSXV Level 1 TMX QuantumFeed.

Each Business message consists of a message header followed by the message body. For a description of the message header's structure, refer to *TMX eXtreme Message Transfer Protocol Specification*. Only the message body's structure is provided in the following sections.

Note: The size of the header is mentioned for message structure completion purposes.

3.1 Message structure

All Business messages must adhere to the Quantum Message Format, as described in *TMX QuantumFeed Service Access Guide*. Following are the structures of different types of Business messages that are disseminated on the TMX Level 1 TMX QuantumFeed.

3.2 Symbol Status

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Stock Group	1	1-Byte Binary	
CUSIP	12	Alphanumeric	
Board Lot	2	2-Byte Binary	
Currency	1	Alphanumeric	
Face Value	8	8-Byte Binary	
Last Sale	8	8-Byte Binary	
Listing Market	1	Alphanumeric	
Product Type	1	Alphanumeric	
Buy Maximum Quantity	4	4-Byte Binary	
Buy Minimum Quantity	4	4-Byte Binary	
Sell Maximum Quantity	4	4-Byte Binary	
Sell Minimum Quantity	4	4-Byte Binary	
Stock State	2	Alphanumeric	
Test Symbol	1	Alphanumeric	
MOC Eligible	1	Alphanumeric	

Total Message Size = 78 bytes

3.3 Trade

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Price	8	8-Byte Binary	
Volume	4	4-Byte Binary	
Buy Broker Number	2	2-Byte Binary	Zero for TSX NAVex™ trades
Sell Broker Number	2	2-Byte Binary	Zero for TSX NAVex trades
Bypass	1	Alphanumeric	
Trade Time Stamp	4	4-Byte Binary	
Settlement Terms	1	Alphanumeric	
Cross Type	1	Alphanumeric	
Last Sale Price	8	8-Byte Binary	
Opening Trade	1	Alphanumeric	
Listing Market	1	Alphanumeric	
Product Type	1	Alphanumeric	
Trade Number	4	4-Byte Binary	
Trade Date	4	4-Byte Binary	
Is Mid Only	1	Alphanumeric	
Is Dark	1	Alphanumeric	
Is Conditional	1	Alphanumeric	

Total Message Size = 69 bytes

3.4 Trade Cancelled

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Volume	4	4-Byte Binary	
Price	8	8-Byte Binary	
Buy Broker Number	2	2-Byte Binary	Zero for TSX NAVex trades
Sell Broker Number	2	2-Byte Binary	Zero for TSX NAVex trades
Trade Time Stamp	4	4-Byte Binary	
Last Sale Price	8	8-Byte Binary	
Listing Market	1	Alphanumeric	
Product Type	1	Alphanumeric	
Trade Number	4	4-Byte Binary	
Trade Date	4	4-Byte Binary	

Total Message Size = 62 bytes

3.5 MOC Imbalance

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Imbalance Side	1	Alphanumeric	
Imbalance Volume	4	4-Byte Binary	
Trading System Time Stamp	8	8-Byte Binary	
Imbalance Reference Price	8	8-Byte Binary	
Paired Volume	4	4-Byte Binary	This field will not be applicable for closing delayed stocks at 4:00p.m.
Market Order Imbalance Volume	4	4-Byte Binary	This field will not be applicable for closing delayed stocks at 4:00p.m.
Market Order Imbalance Side	1	Alphanumeric	This field will not be applicable for closing delayed stocks at 4:00p.m.
Near Indicative Closing Price	8	8-Byte Binary	This field will not be applicable for closing delayed stocks at 4:00p.m.
Far Indicative Closing Price	8	8-Byte Binary	This field will not be applicable for closing delayed stocks at 4:00p.m.
Price Variation	4	4-Byte Binary	This field will not be applicable for closing delayed stocks at 4:00p.m.

Total Message Size = 74 bytes

3.6 Stock Status

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Comment	40	Alphanumeric	
Stock State	2	Alphanumeric	
Trading System Time Stamp	8	8-Byte Binary	
Calculated Closing Price	8	8-Byte Binary	

VWAP	8	8-Byte Binary	
Resume Trade Time	4	4-Byte Binary	
Listing Market	1	Alphanumeric	
Product Type	1	Alphanumeric	

Total Message Size = 96 bytes

3.7 Quote

Field Name	Length (In Bytes)	Data Type	Comments
Message Header	12		
Symbol	12	Alphanumeric	
Bid Price	8	8-Byte Binary	
Bid Size	4	4-Byte Binary	
Ask Price	8	8-Byte Binary	
Ask Size	4	4-Byte Binary	

Total Message Size = 48 bytes

4 Field Definitions

A

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Ask Price	Contains the ask price for the symbol	Binary (8 Bytes)	6 implied decimal places Example: An Ask Price of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the Ask Price.
Ask Size	Contains the ask size volume for the symbol	Binary (4 Bytes)	No implied decimal places Example: An Ask Size of 2500 is represented by 11000100 00001001 00000000 00000000.

B

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Bid Price	Contains the bid price for the symbol	Binary (8 Bytes)	6 implied decimal places Example: A Bid Price of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the Bid Price.
Bid Size	Contains the bid size volume for the symbol	Binary (4 Bytes)	No implied decimal places. Example: A Bid Size of 2500 is represented by 11000100 00001001 00000000 00000000.

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Board Lot	Contains the board lot size	Binary (2 Bytes)	No implied decimal places Example: A Board Lot value of 500 is represented by 11110100 00000001.
Buy Broker Number	An Exchange-assigned public number uniquely identifying a Participating Organization associated with a buy order	Binary (2 Bytes)	No implied decimal places Example: A TSX broker number 009 is represented by 00001001 00000000.
Buy Maximum Quantity	The maximum quantity of a mutual fund allowed to purchase.	Binary (4 Bytes)	No implied decimal places Example: A quantity of 2500 is represented by 11000100 00001001 00000000 00000000.
Buy Minimum Quantity	The minimum quantity of a mutual fund required to purchase.	Binary (4 Bytes)	No implied decimal places Example: A quantity of 2500 is represented by 11000100 00001001 00000000 00000000.
Bypass	An indicator for bypass crosses, which can be printed outside of the Protected NBBO and are not subject to cross interference.	Alphanumeric (1 Byte)	Valid values are: “Y”: The order is a Bypass. “N”: The order is not a Bypass.

C

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Calculated Closing Price	The price at which MOC orders will trade at Closing. It is published only when outside acceptable limits set by the Exchange, triggering an extension to the MOC limit order entry session.	Binary (8 Bytes)	6 implied decimal places Example: A Calculated Closing Price of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the Calculated Closing Price.

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Comment	A text field corresponding to a reason code entered when a stock is halted; or, the initiator of a delayed opening on a stock, or when there is a change to the RT/Oddlot trader on a stock. As well, this is a system-produced text field to describe the disabling of the MOC session by TSX Trading Services.	Alphanumeric (40 Bytes)	
Cross Type	Type of crosses originating from a Participating Organization between managed accounts that have the same manager	Alphanumeric (1 Byte)	Valid values are: • "I": Internal • "B": Basis • "C": Contingent • "D": Derivative-Related • "R": Regular • "S": Special Trading session • "V": VWAP – Volume Weighted Average Price • " ": Default, trade was not a cross.
Currency	The currency associated with a reported price	Alphanumeric (1 Byte)	Valid values are: • "U": USD • "C": CAD
CUSIP	Clearing and settlement registration number	Alphanumeric (12 Bytes)	

F

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Face Value	The face value of a debenture	Binary (8 Bytes)	6 implied decimal places Example: A Face Value of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the face value.
Far Indicative Closing Price	The calculated closing price that will maximize the number of shares matched based on closing interest only (MOC, LOC). This calculation excludes continuous market orders.	Binary (8 Bytes)	See Price.

I

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Imbalance Reference Price	The price that establishes if a given limit-priced Market on Close (LOC) order can contribute to the MOC Imbalance. All LOC orders that are priced equal to or more aggressively than this price will contribute to the MOC Imbalance.	Binary (8 bytes)	See Price.
Imbalance Side	Indicates which side has an imbalance volume for Market on Close	Alphanumeric (1 Byte)	Valid values are: • "B": Buy Side • "S": Sell Side • " ": No imbalance exists
Imbalance Volume	Identifies the volume of shares of the imbalance	Binary (4 Bytes)	No implied decimal places

Field Name	Description	Data Type and Size	Valid Field Values and Comments
	side for Market on Close		Example: A volume of 2,500 is represented by 1100010000001001 0000000000000000.
Is Dark	Indicates the passive side of the trade is dark	Alphanumeric (1 Byte)	"Y" "N"; default is "N"
Is Conditional	Indicates a firm-up conditional to firm-up conditional execution	Alphanumeric (1 Byte)	"Y" "N"; default is "N"

Is Mid Only	Indicates a dark Contra Midpoint Only Plus to Contra Midpoint Only Plus execution	Alphanumeric (1 Byte)	"Y" "N"; default is "N"
-------------	---	-----------------------	---------------------------

L

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Last Sale	Last sale price of a stock	Binary (8 Bytes)	6 implied decimal places Example: A last sale price of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the last sale price.
Listing Market	The market on which a security is listed.	Alphanumeric (1 Byte)	Valid values are: • "T" = TSX • "V" = TSX Venture • "N" = TSX NAVex • "Q" = Nasdaq • "S" = NYSE • "M" = NYSE MKT • "X" = No market (unlisted securities, posted for trading only)

M

Field Name	Description	Data Type and Size	Valid Field Values and Comments
MOC Eligible	Identifies whether the stock is eligible to trade in the Market on Close session each trading day.	Alphanumeric (1 Byte)	Valid values are: • "Y" • "N"
Market Order Imbalance Volume	Indicates the share imbalance when considering MOC orders only. Note this will not change from the	Binary (4 Bytes)	No implied decimal places Example: A volume of 2,500 is represented in binary as 11000100 00001001 00000000 00000000.

	MOC Freeze period as MOC orders are not allowed after this time.		
Market Order Imbalance Side	Side (buy or sell) of the Market Order Imbalance Volume.	Alphanumeric (1 Byte)	Valid Values are: • "B": Buy Side • "S": Sell Side " ": No imbalance exists

N

Field Name	Description	Data Type and Size	Valid Field Values & Comments
Near Indicative Closing Price	The calculated closing price that will maximize the number of shares matched based on on-close orders (MOC, LOC) and visible continuous market orders. Effectively, this is the price at which the closing print would occur at the time of publication.	Binary (8 Bytes)	• See Price.

O

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Opening Trade	Indicates whether the trade occurred at the opening or was the first board lot trade of the day. Note that this field is set to Y on auction trades that occur when a symbol is un-halted and on closing trades.	Alphanumeric (1 Byte)	Valid values are: • “Y” • “N”

P

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Paired Volume	The number of MOC and LOC shares that can be matched at the Reference Price.	Binary (4 Bytes)	No implied decimal places Example: A volume of 2,500 is represented by 11000100 00001001 00000000 00000000.
Price	The price in a valid currency	Binary (8 Bytes)	6 implied decimal places. Example: A Price of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the price.

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Price Variation	This field indicates the absolute value of the percentage of deviation of the Near Indicative Closing Price from the Reference Price.	Binary (4 Bytes)	2 implied decimal places Example 1: A Price Variation of 25.63 percent is represented in binary as 00000011 00001010 00000000 00000000. Example 2: A Price Variation of 125.63 percent is represented in binary as 00010011 00110001 00000000 00000000. The feed client must convert the binary value to decimal and divide the resulting number by 100 to obtain the actual Price Variation value. Note: The decimal value will be 999.99 if the calculated decimal value exceeds 999.99.
Product Type	The product type for a symbol.	Alphanumeric (1 Byte)	Valid values are: • “B” = Debenture • “E” = Equity • “M” = Mutual Fund • “F” = ETF • “U” = US Equity • “O” = Bond

R

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Resume Trade Time	The time at which a halted stock will resume trading	Binary (4 Bytes)	No implied decimal places. Example: A Resume Trade Time of 121010 is represented by 10001000 10100101 10111000 00000000. Feed clients can obtain the Resume Trade Time by converting the binary to decimal. The Resume Trade Time has the format HHMMSSss.

S

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Sell Broker Number	An Exchange-assigned number uniquely identifying a Participating Organization associated with a sell order	Binary (2 Bytes)	No implied decimal places Example: A TSX broker number 009 is represented by 00001001 00000000.
Sell Maximum Quantity	The maximum quantity of a mutual fund allowed to sell.	Binary (4 Bytes)	No implied decimal places Example: A quantity of 2500 is represented by 11000100 00001001 00000000 00000000.
Sell Minimum Quantity	The minimum quantity of a mutual fund required to sell.	Binary (4 Bytes)	No implied decimal places Example: A quantity of 2500 is represented by 11000100 00001001 00000000 00000000.
Settlement Terms	Settlement terms associated with the order	Alphanumeric (1 Byte)	Valid values are: • “C”: Cash • “N”: NN • “M”: MS • “T”: CT • “D”: If there is a valid Settlement Date associated with the order • “ ”: No Settlement Terms

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Stock Group	An identifier of the stock group	Binary (1 Byte)	No implied decimal places Example: A Stock Group of 3 is represented by 00000011.
Stock State	The state of the stock	Alphanumeric (2 Bytes)	Valid values are: • “AR”: Authorized Delayed • “IR”: Inhibited Delayed • “AS”: Authorized Halted • “IS”: Inhibited Halted • “AE”: Authorized Price Movement Delayed • “IE”: Inhibited Price Movement Delayed • “A”: Authorized • “I”: Inhibited
Symbol	A unique alphanumeric identifier for a security	Alphanumeric (12 Bytes)	

T

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Test Symbol	Indicates the symbol is a test symbol.	Alphanumeric (1 Byte)	Valid values are: • “Y” • “N”
Trade Date	Effective date of the trade. Will be filled only when the effective date is different from the current date.	Binary (4 Bytes)	Example: A Trade Date of 20170101 is represented by 01110101 11000101 00110011 00000001. Feed clients can obtain the Trade Date by converting the binary to decimal. The Trade Date format is YYYYMMDD.
Trade Number	A unique identifier per stock assigned to each trade	Binary (4 Bytes)	

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Trade Time Stamp	The time at which the trade occurred, manually set when a trade is added	Binary (4 Bytes)	No implied decimal places Example: A Trade Time Stamp of 121010 is represented by 10110010 11011000 00000001 00000000. Feed clients can obtain the Trade Time Stamp by converting the binary to decimal. The Trade Time Stamp is of the format HHMMSS.
Trading System Time Stamp	The time at which the event being reported occurred in the Trading Engine	Binary (8 Bytes)	Nanoseconds since the Epoch (00:00:00 UTC, January 1, 1970) Example: A Trading System Time Stamp of 2015-07-31 10:14:58.496307008 is represented as 01000000 01011111 11001111 10011100 00001101 11110110 00010011.

V

Field Name	Description	Data Type and Size	Valid Field Values and Comments
Volume	The quantity of shares for an order or trade	Binary (4 Bytes)	No implied decimal places Example: A volume of 2500 is represented by 11000100 00001001 00000000 00000000.
VWAP	Volume weighted average price based on trades occurring in the continuous market for MOC	Binary (8 Bytes)	6 implied decimal places Example: A VWAP of 50.45 is represented by 01010000 11001110 00000001 00000011 00000000 00000000 00000000 00000000. The feed client must convert the binary to decimal and divide the resulting number by 1,000,000 to obtain the VWAP

Appendix A Acronyms and Definitions

- **ASCII**: American Standard Code for Information Interchange
- **COP**: Calculated Opening Price
- **CUSIP**: Committee on Uniform Security Identification Procedures
- **MOC**: Market On Close
- **RT**: Registered Trader
- **TSX**: Toronto Stock Exchange
- **TSXV** : TSX Venture Exchange
- **USD**: United States dollar
- **VWAP**: Volume Weighted Average Price

Appendix B Revision History

Version	Date	Changes
2.6.0	2021/08/10	In 'Business Message Formats - Trade' and "Field Definitions", added a new field 'Is Conditional'.
2.5.2	2021/5/19	In 'Business Message Formats - MOC Imbalance': Added the 'Trading System Time Stamp' field.
2.5.1	2021/04/12	In 'Business Message Formats - MOC Imbalance' and in "Field Definitions": Changed the field name 'Price Variation Indicator' to 'Price Variation'.
2.5.0	2021/03/22	In 'Business Message Types – MOC Imbalance', updated message start time to 3:50p.m. and message frequency to every 10 seconds. In 'Business Message Formats - MOC Imbalance' and in "Field Definitions" added the following 6 new fields: - Paired Volume - Market Order Imbalance Volume - Market Order Imbalance Side - Near Indicative Closing Price - Far Indicative Closing Price - Price Variation Indicator
2.4.0	2020/10/20	Changed length of 'Symbol' from 9 to 12 bytes for Business Message Formats and increased corresponding message sizes. In "Field Definitions", updated size of Symbol from 9 to 12 bytes.
2.3.0	2020/05/21	In "Field Definitions", for Listing Market, added the value "X". In "Field Definitions", for Product Type, added the value "O".
2.2.1	2019/07/22	In "Field Definitions", updated the definition of Is Mid Only
2.2.0	2019/04/16	Added the field "Is Dark" to the Trade message In "Field Definitions", added Is Dark.
2.1.0	2018/01/31	Added the field "Is Mid Only" to the Trade message. In "Field Definitions", added Is Mid Only.
2.0.0	2017/05/08	In "Field Definitions", updated the definition of Bypass. In "Field Definitions", for Cross Type, added the values "D" and " ". In "Field Definitions", for Settlement Terms, added the value " ".
1.50	2017/02/07	In "Field Definitions", added values "Q" = Nasdaq, "S" = NYSE, "M" = NYSE MKT to Listing Market. In "Field Definitions", added the value "U" = US Equity to Product Type.

Version	Date	Changes
1.40	2016/11/25	Added the MOC Eligible field to the symbol Status message. Added the Trade Date field to the Trade and Trade Cancelled messages. In "Field Definitions", added the fields MOC Eligible and Trade Date.
1.30	2016/08/03	In "Business message types", for MOC Imbalance message, added "and again for closing delayed stocks". Added the Stock State and Test Symbol tags to the Symbol Status message. Added the Trade Number field to the Trade and Trade Cancelled messages. In "Field Definitions", added the value "R" to Cross Type. In "Field Definitions", added the value "F" to Product Type. In "Field Definitions", removed the following values from Stock State: "AG" Authorized Frozen, "IG" Inhibited Frozen, "AF" Authorized Price Movement Frozen, and "IF" Inhibited Price Movement Frozen. In "Field Definitions", added the tags Test Symbol and Trade Number.
1.20	2016/03/30	Renamed the Equity Trade message to Trade, the Equity Trade Cancelled message to Trade Cancelled, and the Equity Quote message to Quote. In Business Message Types, added mutual fund to Stock Status message description. In the Trade and Trade Cancelled messages, noted that Buy Broker Number and Sell Broker Number will be zero for TSX NAVex trades. Added Listing Market and Product Type to the Symbol Status, Trade, Trade Cancelled, and Stock Status messages. Added Buy Maximum Quantity, Buy Minimum Quantity, Sell Maximum Quantity, and Sell Minimum Quantity to the Symbol Status message. Added Buy Maximum Quantity, Buy Minimum Quantity, Listing Market, Product Type, Sell Maximum Quantity, and Sell Minimum Quantity to Field Definitions.
1.13	2015/08/11	Modified the definition of Trading System Time Stamp to indicate that it supports nanoseconds.
1.12	2015/04/30	Revised Field Definition for Cross Type to remove 'TSX only' from flags.
1.11	2015/03/20	Added note to definition of Opening Trade field.
1.10	2014/06/26	Added Imbalance Reference Price to MOC Imbalance message and Field Definitions.
1.01	2013/04/08	Corrected example for Resume Trade Time field.
1.0	2010/11/09	Initial release.



TMX Market Operations phone: 416.947.4705
300 – 100 Adelaide St. W. email: marketops@tmx.com
Toronto, ON
M5H 1S3